US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12404877
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Samudrala; Omprakash et al.

Pressure exchangers with fouling and particle handling capabilities

Abstract

A system includes a pressure exchanger and one or more valves. The pressure exchanger includes a rotor that is configured to exchange pressure between a first fluid and a second fluid. The pressure exchanger further includes a housing disposed around the rotor, one or more flushing inlets coupled to the housing, and one or more flushing outlets coupled to the housing. The one or more first valves are coupled to the one or more flushing outlets. The one or more first valves in an open position are associated with a flushing operation. The one or more first valves in a closed position are associated with a lubrication operation.

Inventors: **Samudrala; Omprakash (San Ramon, CA), Baharuddin; Raja Faruq Raja (Castro Valley, CA), Theodossiou; Alexander Patrick (Alameda, CA), Tran; Dung Le (Castro Valley, CA), Gudigopuram; Sujan Reddy (San Ramon, CA)**

Applicant: **Energy Recovery, Inc.** (San Leandro, CA)

Family ID: **86052317**

Assignee: **Energy Recovery, Inc. (San Leandro, CA)**

Appl. No.: **18/849446**

Filed (or PCT Filed): **March 24, 2023**

PCT No.: **PCT/US2023/016296**

PCT Pub. No.: **WO2023/183612**

PCT Pub. Date: September 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250188957 A1	Jun. 12, 2025

Related U.S. Application Data

us-provisional-application US 63323462 20220324

Publication Classification

Int. Cl.: **F04F13/00** (20090101); **F04B53/04** (20060101); **F04B53/06** (20060101); **F04B53/18** (20060101); **F16N31/00** (20060101); F16N33/00 (20060101)

U.S. Cl.:

CPC **F04F13/00** (20130101); **F04B53/04** (20130101); **F04B53/06** (20130101); **F04B53/18** (20130101); **F16N31/00** (20130101); F16N2033/005 (20130101)

Field of Classification Search

CPC: F04B (53/04); F04B (53/06); F04B (53/18); F04F (13/00); F16N (31/00); F16N (2033/005)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2800120	12/1956	Jendrassik et al.	N/A	N/A
2015/0184502	12/2014	Krish	166/177.5	F04F 13/00
2016/0160890	12/2015	Anderson	92/61	F15B 15/063
2018/0087364	12/2017	Krish et al.	N/A	N/A
2018/0347601	12/2017	Hoffman	N/A	F04B 53/20
2023/0008069	12/2022	McLean, Jr.	N/A	F04F 13/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3090187	12/2015	EP	N/A
3635260	12/2019	EP	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for International Application No. PCT/US23/16296, dated Jul. 13, 2023, 16 pages. cited by applicant

Primary Examiner: Freay; Charles G

Attorney, Agent or Firm: Lowenstein Sandler LLP

Background/Summary

RELATED APPLICATIONS (1) This is a national stage application under 35 U.S.C. 371 of International Application PCT/US23/16296, filed Mar. 24, 2023, which claims benefit of U.S.

Provisional Application No. 63/323,462, filed Mar. 24, 2022, the contents of which are incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to pressure exchangers, and, more particularly, pressure exchangers with fouling and particle handling capabilities.

BACKGROUND

(2) Systems use fluids at different pressures. Systems use components to increase pressure of fluid.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is illustrated by way of example, and not by way of limitation in the figures of the accompanying drawings.

(2) FIGS. **1**A-D illustrate schematic diagrams of fluid handling systems including hydraulic energy transfer systems, according to certain embodiments.

(3) FIGS. **2**A-E are exploded perspective views of pressure exchangers (PXs), according to certain embodiments.

(4) FIGS. **3**A-V illustrate PXs, according to certain embodiments.

(5) FIGS. **4**A-D are flow diagram illustrating methods associated with pressure exchanger systems, according to some embodiments.

(6) FIG. **5** is a block diagram illustrating a computer system, according to certain embodiments.

DETAILED DESCRIPTION OF EMBODIMENTS

(7) Embodiments described herein are related to pressure exchangers with fouling and particle handling capabilities.

(8) Systems may use fluids at different pressures. A supply of a fluid to a system may be at lower pressure and one or more portions of the system may operate at higher pressures. A system may include a closed loop with various fluid pressures maintained in different portions of the loop. These systems may include hydraulic fracturing (e.g., fracking or fracing) systems, desalinization systems, refrigeration systems, heat pump systems, energy generation systems, mud pumping systems, slurry pumping systems, industrial fluid systems, waste fluid systems, fluid transportation systems, etc. Pumps or compressors may be used to increase pressure of fluids of such systems.

(9) Conventionally, systems (e.g., refrigeration systems, heat pump systems, reversible heat pump systems, water systems, or the like) use pumps or compressors to increase the pressure of a fluid (e.g., a refrigeration fluid such as carbon dioxide (CO.sub.2), R-744, R-134a, hydrocarbons, hydrofluorocarbons (HFCs), hydrochlorofluorocarbons (HCFCs), ammonia (NH.sub.3), refrigerant blends, R-407A, R-404A, etc.). Conventionally, separate pumps or compressors mechanically coupled to motors are used to increase pressure of the fluid in any portion of a system including an increase in fluid pressure. Pumps and compressors, especially those that operate over a large pressure differential (e.g., cause a large pressure increase in the fluid), require large quantities of energy. Conventional systems thus expend large amounts of energy increasing the pressure of the fluid (via the pumps or compressors driven by the motors). Additionally, conventional fluid transfer systems decrease the pressure of the fluid through expansion valves. Conventional systems inefficiently increase pressure of fluid and decrease pressure of the fluid (e.g., when operating in a loop). This is wasteful in terms of energy used to run the conventional systems (e.g., energy used to repeatedly increase the pressure of the refrigeration fluid to cause increase or decrease of temperature of the surrounding environment).

(10) Conventionally, systems (e.g., refrigeration systems, heat pump systems, reversible heat pump systems, water systems, systems used for reverse osmosis (RO) based industrial waste water treatment plants, or the like) have devices that stall frequently due to fouling (e.g., scaling, organic

growth) and particles becoming embedded in gaps. The devices are then disassembled, cleaned, and reassembled which causes downtime and loss of production. Also, disassembly and reassembly can cause potential damage to components, causing performance degradation or even loss of function. Pre-treatment of fluids (e.g., industrial waste water) to attempt to mitigate fouling and stalling is a challenge as fluids (e.g., water chemistry and composition) can vary (e.g., from plant to plant and from industry to industry). Even if a pre-treatment process is used, there can still be stalling, component damage, downtime, loss of production, etc. to varying in composition of fluids.

(11) The systems, devices, and methods of the present disclosure provide solutions to these and other shortcomings of conventional systems. The present disclosure provides PXs for use in systems (e.g., fluid handling systems, heat transfer systems, refrigeration systems, heat pump systems, cooling systems, heating systems, etc.). In a system, a PX may be configured to exchange pressure between a first fluid (e.g., a high pressure fluid) and a second fluid (e.g., a low pressure fluid). The PX may receive the first fluid via a first inlet (e.g., a high pressure inlet) and a second fluid via a second inlet (e.g., a low pressure inlet). When entering the PX, the first fluid may be of a higher pressure than the second fluid. The PX may exchange pressure between the first fluid and the second fluid. The first fluid may exit the PX via a first outlet (e.g., a low pressure outlet) and the second fluid may exit the PX via a second outlet (e.g., a high pressure outlet). When exiting the PX, the second fluid may have a higher pressure than the first fluid (e.g., pressure has been exchanged between the first fluid and the second fluid).

(12) In some embodiments, a system includes PX includes a rotor configured to exchange pressure between a first fluid and a second fluid. The PX may further include a housing disposed around the rotor, one or more flushing inlets coupled to the housing, and one or more flushing outlets coupled to the housing. One or more first valves may be coupled (e.g., fluidly coupled) to the one or more flushing outlets. The one or more first valves in an open position are associated with a flushing operation (e.g., flushing fluid enters radial bearing gaps via the flushing inlets and exits the circumferential groove via the flushing outlets). The flushing operation may occur when the PX is not exchanging pressure between fluids. The flushing operation may clear particles from the radial bearing gaps and the circumferential grooves. The one or more first valves in a closed position are associated with a lubrication operation (e.g., lubrication fluid enters the radial bearing gaps and exits with the first fluid or the second fluid exiting the PX).

(13) Systems, devices, and methods of the present disclosure provide advantages over conventional solutions. Systems of the present disclosure reduce energy consumption compared to conventional systems. For example, use of a PX of the present disclosure may recover energy stored as pressure and transfer that energy back into the system, reducing the energy cost of operating the system. Systems of the present disclosure may reduce wear on components (e.g., pumps, compressors) compared to conventional systems. Systems of the present disclosure may stall less, have less fouling, have less particles embedded in gaps, have less disassembly cleaning processes, have less downtime, have less component damage, and have less loss of production compared to conventional systems.

(14) Although some embodiments of the present disclosure are described in relation to pressure exchangers, energy recovery devices, and hydraulic energy transfer systems, the current disclosure can be applied to other systems and devices (e.g., pressure exchanger that is not isobaric, rotating components that are not a pressure exchanger, a pressure exchanger that is not rotary, systems that do not include pressure exchangers, etc.).

(15) Although some embodiments of the present disclosure are described in relation to exchanging pressure between fluid used in fracing systems, desalinization systems, heat pump systems, and/or refrigeration systems, the present disclosure can be applied to other types of systems. Fluids can refer to liquid, gas, transcritical fluid, supercritical fluid, subcritical fluid, and/or combinations thereof.

(16) FIGS. **1**A-D illustrate schematic diagrams of fluid handling systems **100** including hydraulic

energy transfer systems **110**, according to certain embodiments.
(17) In some embodiments, a hydraulic energy transfer system **110** includes a pressure exchanger (e.g., PX). The PX may include one or more of the features described in one or more of FIGS. **3**A-V (e.g., inlets, outlets, and valves to perform flushing and lubrication operations) to provide fouling and particle handling capabilities.
(18) The hydraulic energy transfer system **110** (e.g., PX) receives low pressure (LP) fluid in **120** (e.g., low-pressure inlet stream) from a LP in system **122**. The hydraulic energy transfer system **110** also receives high pressure (HP) fluid in **130** (e.g., high-pressure inlet stream) from HP in system **132**. The hydraulic energy transfer system **110** (e.g., PX) exchanges pressure between the HP fluid in **130** and the LP fluid in **120** to provide LP fluid out **140** (e.g., low-pressure outlet stream) to LP fluid out system **142** and to provide HP fluid out **150** (e.g., high-pressure outlet stream) to HP fluid out system **152**.
(19) In some embodiments, the hydraulic energy transfer system **110** includes a PX to exchange pressure between the HP fluid in **130** and the LP fluid in **120**. The PX may be a device that transfers fluid pressure between HP fluid in **130** and LP fluid in **120** at efficiencies in excess of approximately 50%, 60%, 70%, 80%, 90%, or greater (e.g., without utilizing centrifugal technology). High pressure (e.g., HP fluid in **130**, HP fluid out **150**) refers to pressures greater than the low pressure (e.g., LP fluid in **120**, LP fluid out **140**). LP fluid in **120** of the PX may be pressurized and exit the PX at high pressure (e.g., HP fluid out **150**, at a pressure greater than that of LP fluid in **120**), and HP fluid in **130** may be depressurized and exit the PX at low pressure (e.g., LP fluid out **140**, at a pressure less than that of the HP fluid in **130**). The PX may operate with the HP fluid in **130** directly applying a force to pressurize the LP fluid in **120**, with or without a fluid separator between the fluids. Examples of fluid separators that may be used with the PX include, but are not limited to, pistons, bladders, diaphragms, and the like. In some embodiments, PXs may be rotary devices. Rotary PXs, such as those manufactured by Energy Recovery, Inc. of San Leandro, Calif., may not have any separate valves, since the effective valving action is accomplished internal to the device via the relative motion of a rotor with respect to end covers. Rotary PXs may be designed to operate with internal pistons to isolate fluids and transfer pressure with relatively little mixing of the inlet fluid streams. Reciprocating PXs may include a piston moving back and forth in a cylinder for transferring pressure between the fluid streams. Any PX or multiple PXs may be used in the present disclosure, such as, but not limited to, rotary PXs, reciprocating PXs, or any combination thereof. In addition, the PX may be disposed on a skid separate from the other components of a fluid handling system **100** (e.g., in situations in which the PX is added to an existing fluid handling system).
(20) In some embodiments, a motor **160** is coupled to hydraulic energy transfer system **110** (e.g., to a PX). In some embodiments, the motor **160** controls the speed of a rotor of the hydraulic energy transfer system **110** (e.g., to increase pressure of HP fluid out **150**, to decrease pressure of LP fluid out **140**, etc.). In some embodiments, motor **160** generates energy (e.g., acts as a generator) based on pressure exchanging in hydraulic energy transfer system **110**.
(21) The hydraulic energy transfer system **110** may be a hydraulic protection system (e.g., hydraulic buffer system, hydraulic isolation system) that may block or limit contact between solid particle laden fluid (e.g., frac fluid) and various equipment (e.g., hydraulic fracturing equipment, high-pressure pumps) while exchanging work and/or pressure with another fluid. By blocking or limiting contact between various equipment (e.g., fracturing equipment) and solid particle containing fluid, the hydraulic energy transfer system **110** increases the life and performance, while reducing abrasion and wear, of various equipment (e.g., fracturing equipment, high pressure fluid pumps). Less expensive equipment may be used in the fluid handling system **100** by using equipment (e.g., high pressure fluid pumps) not designed for abrasive fluids (e.g., frac fluids and/or corrosive fluids).
(22) The hydraulic energy transfer system **110** may include a hydraulic turbocharger or hydraulic

pressure exchange system, such as a rotating PX. The PX may include one or more chambers (e.g., 1 to 100) to facilitate pressure transfer and equalization of pressures between volumes of first and second fluids (e.g., gas, liquid, multi-phase fluid). In some embodiments, the PX may transfer pressure between a first fluid (e.g., pressure exchange fluid, such as a proppant free or substantially proppant free fluid) and a second fluid that may be highly viscous and/or contain solid particles (e.g., frac fluid containing sand, proppant, powders, debris, ceramics). The solid particle fluid causes abrasion and/or erosion of components of the PX, such as the rotor and end covers of the PX. The fluid (e.g., abrasive particles in the fluid) may cause wear to an interface between the rotor and each end cover as the rotor rotates relative to the end covers. Replacing worn components of the PX may be costly.

(23) The hydraulic energy transfer system **110** may be used in different types of systems, such as fracing systems, desalination systems, refrigeration systems, etc.

(24) FIG. **1**A illustrates a schematic diagram of a fluid handling system **100**A including a hydraulic energy transfer system **110**, according to certain embodiments. Fluid handling system **100**A may include a control module **180** that includes one or more controllers **185**.

(25) FIG. **1**B illustrates a schematic diagram of a fluid handling system **100**B including a hydraulic energy transfer system **110**, according to certain embodiments. Fluid handling system **100**B may be a fracing system. In some embodiments, fluid handling system **100**B includes more components, less components, same routing, different routing, and/or the like than that shown in FIG. **1**B.

(26) LP fluid in **120** and HP fluid out **150** may be frac fluid (e.g., fluid including solid particles, proppant fluid, etc.). HP fluid in **130** and LP fluid out **140** may be substantially solid particle free fluid (e.g., proppant free fluid, water, filtered fluid, etc.).

(27) LP in system **122** may include one or more low pressure fluid pumps to provide LP fluid in **120** to the hydraulic energy transfer system **110** (e.g., PX). HP in system **132** may include one or more high pressure fluid pumps **134** to provide HP fluid in **130** to hydraulic energy transfer system **110**.

(28) Hydraulic energy transfer system **110** exchanges pressure between LP fluid in **120** (e.g., low pressure frac fluid) and HP fluid in **130** (e.g., high pressure water) to provide HP fluid out **150** (e.g., high pressure frac fluid) to HP out system **152** and to provide LP fluid out **140** (e.g., low pressure water). HP out system **152** may include a rock formation **154** (e.g., well) that includes cracks **156**. The solid particles (e.g., proppants) from HP fluid out **150** may be provided into the cracks **156** of the rock formation.

(29) In some embodiments, LP fluid out **140**, high pressure fluid pumps **134**, and HP fluid in **130** are part of a first loop (e.g., proppant free fluid loop). The LP fluid out **140** may be provided to the high pressure fluid pumps to generate HP fluid in **130** that becomes LP fluid out **140** upon exiting the hydraulic energy transfer system **110**.

(30) In some embodiments, LP fluid in **120**, HP fluid out **150**, and low pressure fluid pumps **124** are part of a second loop (e.g., proppant containing fluid loop). The HP fluid out **150** may be provided into the rock formation **154** and then pumped from the rock formation **154** by the low pressure fluid pumps **124** to generate LP fluid in **120**.

(31) In some embodiments, fluid handling system **100**B is used in well completion operations in the oil and gas industry to perform hydraulic fracturing (e.g., fracking, fracing) to increase the release of oil and gas in rock formations **154**. HP out system **152** may include rock formations **154** (e.g., a well). Hydraulic fracturing may include pumping HP fluid out **150** containing a combination of water, chemicals, and solid particles (e.g., sand, ceramics, proppant) into a well (e.g., rock formation **154**) at high pressures. LP fluid in **120** and HP fluid out **150** may include a particulate laden fluid that increases the release of oil and gas in rock formations **154** by propagating and increasing the size of cracks **156** in the rock formations **154**. The high pressures of HP fluid out **150** initiates and increases size of cracks **156** and propagation through the rock formation **154** to release more oil and gas, while the solid particles (e.g., powders, debris, etc.)

enter the cracks **156** to keep the cracks **156** open (e.g., prevent the cracks **156** from closing once HP fluid out **150** is depressurized).

(32) In order to pump this particulate laden fluid into the rock formation **154** (e.g., a well), the fluid handling system **100**B may include one or more high pressure fluid pumps **134** and one or more low pressure fluid pumps **124** coupled to the hydraulic energy transfer system **110**. For example, the hydraulic energy transfer system **110** may be a hydraulic turbocharger or a PX (e.g., a rotary PX). In operation, the hydraulic energy transfer system **110** transfers pressures without any substantial mixing between a first fluid (e.g., HP fluid in **130**, proppant free fluid) pumped by the high pressure fluid pumps **134** and a second fluid (e.g., LP fluid in **120**, proppant containing fluid, frac fluid) pumped by the low pressure fluid pumps **124**. In this manner, the hydraulic energy transfer system **110** blocks or limits wear on the high pressure fluid pumps **134**, while enabling the fluid handling system **100**B to pump a high-pressure frac fluid (e.g., HP fluid out **150**) into the rock formation **154** to release oil and gas. In order to operate in corrosive and abrasive environments, the hydraulic energy transfer system **110** may be made from materials resistant to corrosive and abrasive substances in either the first and second fluids. For example, the hydraulic energy transfer system **110** may be made out of ceramics (e.g., alumina, cermets, such as carbide, oxide, nitride, or boride bard phases) within a metal matrix (e.g., Co, Cr or Ni or any combination thereof) such as tungsten carbide in a matrix of CoCr, Ni, NiCr or Co.

(33) In some embodiments, the hydraulic energy transfer system **110** includes a PX (e.g., rotary PX) and HP fluid in **130** (e.g., the first fluid, high-pressure solid particle free fluid) enters a first side of the PX where the HP fluid in **130** contacts LP fluid in **120** (e.g., the second fluid, low-pressure frac fluid) entering the PX on a second side. The contact between the fluids enables the HP fluid in **130** to increase the pressure of the second fluid (e.g., LP fluid in **120**), which drives the second fluid out (e.g., HP fluid out **150**) of the PX and down a well (e.g., rock formation **154**) for fracturing operations. The first fluid (e.g., LP fluid out **140**) similarly exits the PX, but at a low pressure after exchanging pressure with the second fluid. As noted above, the second fluid may be a low-pressure frac fluid that may include abrasive particles, which may wear the interface between the rotor and the respective end covers as the rotor rotates relative to the respective end covers.

(34) FIG. **1**C illustrates a schematic diagram of a fluid handling system **100**C including a hydraulic energy transfer system **110**, according to certain embodiments. Fluid handling system **100**C may be a desalination system (e.g., remove salt and/or other minerals from water). In some embodiments, fluid handling system **100**C includes more components, less components, same routing, different routing, and/or the like than that shown in FIG. **1**C.

(35) LP in system **122** may include a feed pump **126** (e.g., low pressure fluid pump **124**) that receives seawater in **170** (e.g., feed water from a reservoir or directly from the ocean) and provides LP fluid in **120** (e.g., low pressure seawater, feed water) to hydraulic energy transfer system **110** (e.g., PX). HP in system **132** may include membranes **136** that provide HP fluid in **130** (e.g., high pressure brine) to hydraulic energy transfer system **110** (e.g., PX). The hydraulic energy transfer system **110** exchanges pressure between the HP fluid in **130** and LP fluid in **120** to provide HP fluid out **150** (e.g., high pressure seawater) to HP out system **152** and to provide LP fluid out **140** (e.g., low pressure brine) to LP out system **142** (e.g., geological mass, ocean, sea, discarded, etc.).

(36) The membranes **136** may be a membrane separation device configured to separate fluids traversing a membrane, such as a reverse osmosis membrane. Membranes **136** may provide HP fluid in **130** which is a concentrated feed-water or concentrate (e.g., brine) to the hydraulic energy transfer system **110**. Pressure of the HP fluid in **130** may be used to compress low-pressure feed water (e.g., LP fluid in **120**) to be high pressure feed water (e.g., HP fluid out **150**). For simplicity and illustration purposes, the term feed water is used. However, fluids other than water may be used in the hydraulic energy transfer system **110**.

(37) The circulation pump **158** (e.g., centrifugal pump) provides the HP fluid out **150** (e.g., high pressure seawater) to membranes **136**. The membranes **136** filter the HP fluid out **150** to provide

LP potable water **172** and HP fluid in **130** (e.g., high pressure brine). The LP out system **142** provides brine out **174** (e.g., to geological mass, ocean, sea, discarded, etc.).

(38) In some embodiments, a high pressure fluid pump **176** is disposed between the feed pump **126** and the membranes **136**. The high pressure fluid pump **176** increases pressure of the low pressure seawater (e.g., LP fluid in **120**, provides high pressure feed water) to be mixed with the high pressure seawater provided by circulation pump **158**.

(39) In some embodiments, use of the hydraulic energy transfer system **110** decreases the load on high pressure fluid pump **176**. In some embodiments, fluid handling system **100**C provides LP potable water **172** without use of high pressure fluid pump **176**. In some embodiments, fluid handling system **100**C provides LP potable water **172** with intermittent use of high pressure fluid pump **176**.

(40) In some examples, hydraulic energy transfer system **110** (e.g., PX) receives LP fluid in **120** (e.g., low-pressure feed-water) at about **30** pounds per square inch (PSI) and receives HP fluid in **130** (e.g., high-pressure brine or concentrate) at about **980** PSI. The hydraulic energy transfer system **110** (e.g., PX) transfers pressure from the high-pressure concentrate (e.g., HP fluid in **130**) to the low-pressure feed-water (e.g., LP fluid in **120**). The hydraulic energy transfer system **110** (e.g., PX) outputs HP fluid out **150** (e.g., high pressure (compressed) feed-water) at about 965 PSI and LP fluid out **140** (e.g., low-pressure concentrate) at about 15 PSI. Thus, the hydraulic energy transfer system **110** (e.g., PX) may be about 97% efficient since the input volume is about equal to the output volume of the hydraulic energy transfer system **110** (e.g., PX), and 965 PSI is about 97% of 980 PSI.

(41) FIG. **1**D illustrates a schematic diagram of a fluid handling system **100**D including a hydraulic energy transfer system **110**, according to certain embodiments. Fluid handling system **100**D may be a refrigeration system. In some embodiments, fluid handling system **100**D includes more components, less components, same routing, different routing, and/or the like than that shown in FIG. **1**D.

(42) Hydraulic energy transfer system **110** (e.g., PX) may receive LP fluid in **120** from LP in system **122** (e.g., low pressure lift device **128**, low pressure fluid pump, etc.) and HP fluid in **130** from HP in system **132** (e.g., condenser **138**). The hydraulic energy transfer system **110** (e.g., PX) may exchange pressure between the LP fluid in **120** and HP fluid in **130** to provide HP fluid out **150** to HP out system **152** (e.g., high pressure lift device **159**) and to provide LP fluid out **140** to LP out system **142** (e.g., evaporator **144**). The evaporator **144** may provide the fluid to compressor **178** and low pressure lift device **128**. The condenser **138** may receive fluid from compressor **178** and high pressure lift device **159**.

(43) The fluid handling system **100**D may be a closed system. LP fluid in **120**, HP fluid in **130**, LP fluid out **140**, and HP fluid out **150** may all be a fluid (e.g., refrigerant) that is circulated in the closed system of fluid handling system **100**D.

(44) In some embodiments, the fluid of fluid handling system **100**D may include solid particles. For example, the piping, equipment, connections (e.g., pipe welds, pipe soldering), etc. may introduce solid particles (e.g., solid particles from the welds) into the fluid in the fluid handling system **100**D. The solid particles in the fluid and/or the high pressure of the fluid may cause abrasion and/or erosion of components (e.g., rotor, end covers) of the PX of hydraulic energy transfer system **110**.

(45) FIGS. **2**A-E are exploded perspective views a rotary PX **40** (e.g., rotary pressure exchanger, rotary liquid piston compressor (LPC)), according to certain embodiments. PX **40** may include a motor **92** and/or a control module **94**.

(46) In some embodiments, PX **40** includes one or more of the features described in one or more of FIGS. **3**A-V (e.g., inlets, outlets, and valves to perform flushing and lubrication operations) to provide fouling and particle handling capabilities.

(47) PX **40** is configured to transfer pressure and/or work between a first fluid (e.g., proppant free

fluid or supercritical carbon dioxide, HP fluid in **130**) and a second fluid (e.g., frac fluid or superheated gaseous carbon dioxide, LP fluid in **120**) with minimal mixing of the fluids. The rotary PX **40** may include a generally cylindrical body portion **42** that includes a sleeve **44** (e.g., rotor sleeve) and a rotor **46**. The rotary PX **40** may also include two end caps **48** and **50** that include manifolds **52** and **54**, respectively. Manifold **52** includes respective inlet port **56** and outlet port **58**, while manifold **54** includes respective inlet port **60** and outlet port **62**. In operation, these inlet ports **56**, **60** enable the first and second fluids to enter the rotary PX **40** to exchange pressure, while the outlet ports **58**, **62** enable the first and second fluids to then exit the rotary PX **40**. In operation, the inlet port **56** may receive a high-pressure first fluid (e.g., HP fluid in **130**), and after exchanging pressure, the outlet port **58** may be used to route a low-pressure first fluid (e.g., LP fluid out **140**) out of the rotary PX **40**. Similarly, the inlet port **60** may receive a low-pressure second fluid (e.g., LP fluid in **120**) and the outlet port **62** may be used to route a high-pressure second fluid (e.g., HP fluid out **150**) out of the rotary PX **40**. The end caps **48** and **50** include respective end covers **64** and **66** (e.g., end plates) disposed within respective manifolds **52** and **54** that enable fluid sealing contact with the rotor **46**.

(48) As noted above, one or more components of the PX **40**, such as the rotor **46**, the end cover **64**, and/or the end cover **66**, may be constructed from a wear-resistant material (e.g., carbide, cemented carbide, silicon carbide, tungsten carbide, etc.) with a hardness greater than a predetermined threshold (e.g., a Vickers hardness number that is at least 1000, 1250,1500, 1750, 2000, 2250, or more). For example, tungsten carbide may be more durable and may provide improved wear resistance to abrasive fluids as compared to other materials, such as alumina ceramics.

(49) The rotor **46** may be cylindrical and disposed in the sleeve **44**, which enables the rotor **46** to rotate about the axis **68**. The rotor **46** may have a plurality of channels **70** (e.g., ducts, rotor ducts) extending substantially longitudinally through the rotor **46** with openings **72** and **74** (e.g., rotor ports) at each end arranged symmetrically about the longitudinal axis **68**. The openings **72** and **74** of the rotor **46** are arranged for hydraulic communication with inlet and outlet apertures **76** and **78** (e.g., end cover inlet port and end cover outlet port) and **80** and **82** (e.g., end cover inlet port and end cover outlet port) in the end covers **64** and **66**, in such a manner that during rotation the channels **70** are exposed to fluid at high-pressure and fluid at low-pressure. As illustrated, the inlet and outlet apertures **76** and **78** and **80** and **82** may be designed in the form of arcs or segments of a circle (e.g., C-shaped).

(50) In some embodiments, a controller using sensor feedback (e.g., revolutions per minute measured through a tachometer or optical encoder or volume flow rate measured through flowmeter) may control the extent of mixing between the first and second fluids in the rotary PX **40**, which may be used to improve the operability of the fluid handling system (e.g., fluid handling systems **100**A-D of FIGS. **1**A-D). For example, varying the volume flow rates of the first and second fluids entering the rotary PX **40** allows the plant operator (e.g., system operator) to control the amount of fluid mixing within the PX **40**. In addition, varying the rotational speed of the rotor **46** also allows the operator to control mixing. Three characteristics of the rotary PX **40** that affect mixing are: (1) the aspect ratio of the rotor channels **70**; (2) the duration of exposure between the first and second fluids; and (3) the creation of a fluid barrier (e.g., an interface) between the first and second fluids within the rotor channels **70**. First, the rotor channels **70** (e.g., ducts) are generally long and narrow, which stabilizes the flow within the rotary PX **40**. In addition, the first and second fluids may move through the channels **70** in a plug flow regime with minimal axial mixing. Second, in certain embodiments, the speed of the rotor **46** reduces contact between the first and second fluids. For example, the speed of the rotor **46** (e.g., rotor speed of approximately 1200 RPM) may reduce contact times between the first and second fluids to less than approximately 0.15 seconds, 0.10 seconds, or 0.05 seconds. Third, a small portion of the rotor channel **70** is used for the exchange of pressure between the first and second fluids. Therefore, a volume of fluid remains in the channel **70** as a barrier between the first and second fluids. All these mechanisms may limit

mixing within the rotary PX **40**. Moreover, in some embodiments, the rotary PX **40** may be designed to operate with internal pistons or other barriers, either complete or partial, that isolate the first and second fluids while enabling pressure transfer.

(51) FIGS. **2**B-**2**E are exploded views of an embodiment of the rotary PX **40** illustrating the sequence of positions of a single rotor channel **70** in the rotor **46** as the channel **70** rotates through a complete cycle. It is noted that FIGS. **2**B-**2**E are simplifications of the rotary PX **40** showing one rotor channel **70**, and the channel **70** is shown as having a circular cross-sectional shape. In other embodiments, the rotary PX **40** may include a plurality of channels **70** with the same or different cross-sectional shapes (e.g., circular, oval, square, rectangular, polygonal, etc.). Thus, FIGS. **2**B-**2**E are simplifications for purposes of illustration, and other embodiments of the rotary PX **40** may have configurations different from that shown in FIGS. **2**A-**2**E. As described in detail below, the rotary PX **40** facilitates pressure exchange between first and second fluids by enabling the first and second fluids to briefly contact each other within the rotor **46**. In certain embodiments, this exchange happens at speeds that result in limited mixing of the first and second fluids. The speed of the pressure wave traveling through the rotor channel **70** (as soon as the channel is exposed to the aperture **76**), the diffusion speeds of the fluids, and the rotational speed of rotor **46** dictate whether any mixing occurs and to what extent.

(52) FIG. **2**B is an exploded perspective view of an embodiment of a rotary PX **40** (e.g., rotary LPC), according to certain embodiments. In FIG. **2**B, the channel opening **72** is in a first position. In the first position, the channel opening **72** is in fluid communication with the aperture **78** in end cover **64** and therefore with the manifold **52**, while the opposing channel opening **74** is in hydraulic communication with the aperture **82** in end cover **66** and by extension with the manifold **54**. As will be discussed below, the rotor **46** may rotate in the clockwise direction indicated by arrow **84**. In operation, low-pressure second fluid **86** passes through end cover **66** and enters the channel **70**, where it contacts the first fluid **88** at a dynamic fluid interface **90**. The second fluid **86** then drives the first fluid **88** out of the channel **70**, through end cover **64**, and out of the rotary PX **40**. However, because of the short duration of contact, there is minimal mixing between the second fluid **86** and the first fluid **88**.

(53) FIG. **2**C is an exploded perspective view of an embodiment of a rotary PX **40** (e.g., rotary LPC), according to certain embodiments. In FIG. **2**C, the channel **70** has rotated clockwise through an arc of approximately 90 degrees. In this position, the opening **74** (e.g., outlet) is no longer in fluid communication with the apertures **80** and **82** of end cover **66**, and the opening **72** is no longer in fluid communication with the apertures **76** and **78** of end cover **64**. Accordingly, the low-pressure second fluid **86** is temporarily contained within the channel **70**.

(54) FIG. **2**D is an exploded perspective view of an embodiment of a rotary PX **40** (e.g., rotary LPC), according to certain embodiments. In FIG. **2**D, the channel **70** has rotated through approximately 60 degrees of arc from the position shown in FIG. **2**B. The opening **74** is now in fluid communication with aperture **80** in end cover **66**, and the opening **72** of the channel **70** is now in fluid communication with aperture **76** of the end cover **64**. In this position, high-pressure first fluid **88** enters and pressurizes the low-pressure second fluid **86**, driving the second fluid **86** out of the rotor channel **70** and through the aperture **80**.

(55) FIG. **2**E is an exploded perspective view of an embodiment of a rotary PX **40** (e.g., rotary LPC), according to certain embodiments. In FIG. **2**E, the channel **70** has rotated through approximately 270 degrees of arc from the position shown in FIG. **2**B. In this position, the opening **74** is no longer in fluid communication with the apertures **80** and **82** of end cover **66**, and the opening **72** is no longer in fluid communication with the apertures **76** and **78** of end cover **64**. Accordingly, the first fluid **88** is no longer pressurized and is temporarily contained within the channel **70** until the rotor **46** rotates another 90 degrees, starting the cycle over again.

(56) Abrasion and/or erosion damage in a PX may occur when suspended solids are introduced and mixed in the fluid that enters the PX. Abrasion damage may occur when particles enter gaps in the

PX (e.g., trapped between a stationary end cover and a rotating rotor). Erosion damage may occur due to existence of suspended solids (e.g., erodents) in high velocity fluid jets (e.g., slurry jets) that are formed due to the high pressure differentials inside the PX. When the high velocity jet makes an impact with components of the PX, the high velocity jet can cause damage to those components. Damage (e.g., erosion damage) can occur when a high pressure rotor port (e.g., rotor duct) opens to a low pressure end cover port (e.g., kidney) or when a low pressure rotor port (e.g., rotor duct) opens to a high pressure end cover port (e.g., kidney) which causes a high pressure differential.
(57) FIGS. **3**A-V illustrate PXs **300**, according to some embodiments. Hydraulic energy transfer system **110** of one or more of FIGS. **1**A-D and/or PX **40** of one or more of FIGS. **2**A-E may include one or more features, materials, functionalities, etc. that are the same as or similar to those of FIGS. **3**A-V. PXs described in FIGS. **3**A-V may include one or more features, materials, functionalities, etc. as described in relation to one or more of FIGS. **1**A-**2**E.
(58) PX **300** (e.g., rotary pressure exchanger) may have fouling and particle handling capabilities. PX **300** may be used for energy recovery in RO-based industrial waste water treatment plants. Conventional devices stall frequently due to fouling (e.g., scaling, organic growth) and particles getting embedded in gaps (e.g., narrow bearing gaps). The devices are then disassembled, bearing gaps are cleaned, and then the devices are reassembled which causes downtime. Disassembly and reassembly may damage components (e.g., brittle ceramics) which causes performance degradation and loss of function. Pre-treatment of industrial waste water to mitigate fouling and stalling is a challenge as water chemistry and composition varies from plant to plant and from industry to industry. To solve these and other problems, PX **300** (e.g., and system **301**) of the present disclosure has the ability to manage foulants and particles without stalling.
(59) System **301** may have a clean-in-place (CIP) system in the PX **300** which may allow for periodic flushing of the bearing gaps with fluid (e.g., cleaning agent, water, etc.). PX **300** may have features that prevent or delay fouling or accumulation of particles in bearing gaps. System **301** (e.g., clean-in-place system) may provide clean, high-pressure fluid to the portions of the PX **300** (e.g., to bearings, etc.) during operation of the PX **300** (e.g., exchanging of pressure between fluids) to prevent fouling and stalling. In some embodiments, a self-cleaning in-line bearing filter embedded in the PX **300** and can provide filtered fluid to the bearings (or other features of the PX **300**).
(60) FIG. **3**A illustrates side cross-sectional view of a PX **300**, according to certain embodiments. FIG. **3**B illustrates front cross-sectional view of a PX **300**, according to certain embodiments.
(61) PX **300** includes a rotor **310**, end covers **320**A-B, sleeve **330**, housing **340**, end caps **350**A-B, tension rod **360** (e.g., center post, shaft, etc.), and ports **370**A-D. PX **300** may include flushing inlets **342**A-B (e.g., see FIG. **3**A), lubrication outlet **344** (e.g., see FIG. **3**A), and/or flushing outlets **346**A-B (e.g., see FIG. **3**B).
(62) Scale formation may occur in the radial bearing gaps **380** between rotor **310** and sleeve **330**. Deposit or accumulation (e.g., any deposit or accumulation) in the radial bearing gaps **380** (e.g., due to micron level clearances) can cause the rotor **310** to slow down (e.g., enhancing mixing) or stall (e.g., loss of function of PX **300**). Axial bearing gaps **384**, circumferential grooves **382**, outside diameter (OD) of sleeve **330**, and center bore **386** are other locations where fouling can occur either due to small gaps or due to flow velocities being too small.
(63) Flushing inlets **342**A-B and flushing outlets **346**A-B may be fluidly coupled to one or more features (e.g., gaps, grooves, radial bearing gaps **380**, bearing fluid plenum **381**, axial bearing gaps **384**, circumferential grooves **382**, OD of sleeve **330**, center bore **386**, etc.) of the PX **300**.
(64) In some embodiments, flushing inlets **342**A-B are fluidly coupled to radial bearing gap **380** (e.g., via bearing fluid plenum **381**) of PX **300** (e.g., a radial bearing gap **380** may be the two small gap end areas between rotor **310** and sleeve **330**, bearing fluid plenum **381** may be a central area between rotor **310** and sleeve **330** where the rotor has a smaller diameter) and flushing outlets **346**A-B are coupled to circumferential grooves **382** (e.g., circumferential grooves **382** may be

formed by end covers **320** with the rotor **310** and/or sleeve **330** each overlapping a portion of the circumferential grooves **382**). Lubrication outlet **344** may be coupled to a port **370** (e.g., port **370**B, high pressure out (HPOUT) port).

(65) The different operations may be controlled by actuating valves **390** of system **301**. In some embodiments, system **301** may perform one or more operations more frequently (e.g., continuously, substantially continuously, internal lubrication operation during use of PX **300**) and one or more operations less frequently (e.g., flushing operation when PX **300** is not operating. System **301** may perform the operations (e.g., flushing and/or lubrication operations) based on sensor data, based on duration of operation of PX **300**, based on properties of the fluids entering and exiting the PX **300**, and/or the like.

(66) FIGS. **3**C-E illustrate system **301** in an internal lubrication configuration, according to certain embodiments. System **301** (e.g., clean-in-place system) can be used to supply clean high pressure bearing fluid (e.g., internal lubrication operation) during operation of PX **300** (e.g., while PX **300** exchanges pressure between fluids). During operation of PX **300** (e.g., during normal operation), flushing line coming to the PX **300** (e.g., via valve **390**E) and outlet valves **390**A-B may be closed and HPOUT fluid may be taken from HPOUT plenum (e.g., from HPOUT port, from plenum between end cover **320**B and port **370**B) in the PX **300** (e.g., valve **390**C may be in an open position) and fed through external piping to the radial bearing gap **380** (e.g., via bearing fluid plenum **381**).

(67) In some embodiments (e.g., during an internal lubrication operation during operation of PX **300**), lubrication outlet **344** provides fluid flow (e.g., from HPOUT) to one or more flushing inlets **342** (e.g., flushing inlets **342**A-B) which enters (e.g., lubricates) radial bearing gap **380** (e.g., via bearing fluid plenum **381**), then flows to circumferential groove **382**, then flows to axial bearing gaps **384**A-B, and then exits via a port **370** (e.g., low pressure out (LPOUT) port).

(68) FIGS. **3**F-H illustrate system **301** in an external lubrication configuration, according to certain embodiments. System **301** (e.g., clean-in-place system) can be used to supply clean high pressure bearing fluid (e.g., internal lubrication operation) during operation of PX **300** (e.g., while PX **300** exchanges pressure between fluids). During operation of PX **300** (e.g., during normal operation), flushing outlet valves **390**A-B may be closed, HPOUT bearing fluid supply can be shut off (e.g., valve **390**C is in a closed position) and high pressure external clean bearing fluid can be piped in to keep PX **300** running for long times without fouling.

(69) In some embodiments (e.g., during an external lubrication operation during operation of PX **300**), external bearing fluid is provided to one or more flushing inlets **342** (e.g., flushing inlets **342**A-B) which enters (e.g., lubricates) radial bearing gap **380** (e.g., via bearing fluid plenum **381**), then flows to axial bearing gaps **384**A-B, then flows to circumferential groove **382**, and then exits via a port **370** (e.g., LPOUT port).

(70) FIGS. **3**I-K illustrate system **301** in a flushing configuration, according to certain embodiments. Operation of PX **300** is stopped (e.g., once a day) using valves on ports **370**. Flushing system valves **390**A-B are opened (e.g., and valve **390**C is in a closed position) and high pressure flushing fluid (e.g., with appropriate chemicals such as scale dissolving agent, such as citric acid, at the right concentration) is fed (e.g., entering via flushing inlets **342**) through the bearings (e.g., radial bearing gap **380** and/or axial bearing gaps **384**) and out through the passages connected to the circumferential grooves **382** (e.g., exiting via flushing outlets **346**A and **346**B). Flushing pressure, flushing flow rate, flushing time, flushing fluid, and/or flushing interval are selected based on the particular application. In some embodiments, system **301** (e.g., flushing operation) can be used to soak the bearings (e.g., radial bearing gap **380**, bearing fluid plenum **381**, and/or axial bearing gaps **384**) in scale dissolver or biocide (e.g., entering via flushing inlets **342**) to remove fouling (e.g., without continuous flow).

(71) In some embodiments (e.g., during a flushing operation, a clean-in-place operation, when PX **300** is not operating), external cleaning-in-place fluid (e.g., cleaning agent) is provided to one or

more flushing inlets **342** (e.g., flushing inlets **342**A-B) which enters (e.g., flushes) radial bearing gap **380**, then flows to circumferential grooves **382**, and then exits via flushing outlets **346**A-B (e.g., that are fluidly coupled to circumferential grooves **382**).

(72) FIGS. **3**L-N illustrate system **301** in a flushing configuration, according to certain embodiments. In some embodiments (e.g., during a flushing operation during operation of PX **300**), lubrication outlet **344** provides fluid flow (e.g., from HPOUT) to one flushing inlet **342** (e.g., flushing inlet **342**A) which enters (e.g., flushes) radial bearing gap **380** and exits via a different flushing inlet **342** (e.g., flushing inlet **342**B) (e.g., to be drained via drainage line). Valve **390**D (e.g., disposed between flushing inlet **324**A and lubrication outlet **344**) may be in a closed position to enable this operation. Valve **390**C may be in an open position and valves **390**A-B and valve **390**E may be in a closed position.

(73) FIGS. **3**O-P illustrate PXs **300** that have a collection groove **392** (e.g., formed by sleeve **330**), according to certain embodiments.

(74) PX **300** may have one or more features (e.g., collection groove **392**) configured to prevent or delay fouling and particle accumulation in the radial bearing gap **380** and circumferential grooves **382**. In some embodiments, the circumferential grooves **382** are formed by the rotor **310** (e.g., circumferential grooves **382** are not made by the end covers **320**). Circumferential grooves **382** in the rotor **310** may be less prone to fouling due to the rotation of rotor **310**. A collection groove **392** (e.g., long substantially vertical slot) may be machined onto the inner surface of sleeve **330** (e.g., on low pressure side of PX **300**). One or more channels **394** (e.g., thru-holes) may be machined from one of the ducts of rotor **310** to an outer surface of rotor **310**. The channels **394** may connect with collection groove **392** when aligned with the low pressure kidneys of the end cover. Particles in the radial bearing gap **380** (e.g., and bearing fluid plenum **381**) may be collected in the collection groove **392** and may be flushed away into LPOUT fluid flow through the channels **394** (e.g., duct holes). This may occur at least once per revolution (e.g., about once per revolution).

(75) FIGS. **3**Q-R illustrate PXs that have a collection groove **392** (e.g., formed by rotor **310**), according to certain embodiments. Collection groove **392** may be machined onto the rotor. Sleeve **330** may have sleeve passages **393** that connect the collection groove **392** to the LPOUT fluid flow at least once per revolution (e.g., about once per revolution).

(76) FIGS. **3**S-V are associated with a filter **396** (e.g., self-cleaning in-line bearing fluid filter) of PX **300**, according to certain embodiments. FIG. **3**S illustrates a perspective view of a PX **300** including a filter **396**. FIG. **3**T illustrates a perspective view of a filter **396**. FIG. **3**U illustrates a cross-sectional view of a filter **396**. FIG. **3**V illustrates a filter **396** disposed in an end cap **350** of PX **300**.

(77) Referring to FIG. **3**S, fluid flow exiting end cover **320**B (e.g., HPOUT fluid flow) flows through filter **396** and then to filtered fluid supply **399** to one or more features of the PX **300** (e.g., one or more of bearing gap, radial bearing gap **380**, axial bearing gap **384**, circumferential groove **382**, center bore **386**, etc.). In some embodiments, a valve controls filtered fluid flow via filtered fluid supply **399**. A controller **303** (e.g., see FIGS. **3**C-**3**N) may actuate (e.g., periodically, based on sensor data, based on a schedule, etc.) the valve to an open position to provide the filtered fluid supply **399** to one or more features of the PX **300**. In some embodiments, filter **396** filters HPOUT fluid flow to provide via the filtered fluid supply **399**. In some embodiments, filter **396** filters HPIN fluid flow to provide via the filtered fluid supply **399**.

(78) In some embodiments, the filter **396** may prevent greater than about 10 micron suspended solid particles from reaching the bearing gaps (e.g., radial bearing gap **380**, axial bearing gap **384**, etc.). The filter **396** may be made of a porous metal or a hydrophobic porous plastic may be attached to the end cap **350** in the HPOUT plenum. The filter **396** may include a filter element housed in a metal cage that is threaded onto the end cap **350**. HPOUT flow swirling in the plenum may clean the filter **396** and the filtered fluid may be collected (e.g., through a pipe, via a pathway formed by PX **300**, via a pathway formed by sleeve **330** or housing **340**) and fed to (e.g., directly

to) the bearings (e.g., radial bearing gap **380**, axial bearing gap **384**, etc.). Filter **396** may be used with any of the embodiments of the present disclosure.

(79) In some embodiments, system **301** is used in RO-based industrial waste water treatment. System **301** may allow use of PX **300** for energy recovery in RO-based industrial waste water treatment facilities. System **301** may allow use of PX when the process stream contains scale-causing ions, high chemical oxygen demand (COD) and/or biochemical oxygen demand (BOD) content, large concentrations of suspended particles, etc.

(80) The present disclosure (e.g., one or more embodiments of FIGS. **3**A-V) may have improvement in one or more of pressure range, efficiency, reduction in volume and cost, etc. compared to conventional solutions.

(81) In some embodiments, one or more features of PX **300** includes a coating and/or surface treatment for biofouling and/or scaling resistance. The one or more features may include one or more of the circumferential groove **382** of end cover **320**, center bore **386** of rotor **310**, ducts of rotor **310**, outer surface (e.g., outside diameter, lube hole) of sleeve **330**, gap (e.g., radial bearing gap **380**) between rotor **310** and sleeve **330**, axial surface of rotor **310**, and/or the like. One or more features of PX **300** may have reduced surface roughness, texturing to enhance hydrophobicity, altered surface chemical composition, surface charge, anti-adhesion coatings, oleophobic coatings, coatings with silver nano particles (e.g., silver nano particle coating), and/or the like.

(82) FIGS. **4**A-D are flow diagrams illustrating methods **400**A-D associated with pressure exchanger systems (e.g., system **301** and/or PX **300** of one or more of FIGS. **3**A-V), according to certain embodiments. In some embodiments, methods **400**A-D are performed by processing logic that includes hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, processing device, etc.), software (such as instructions run on a processing device, a general purpose computer system, or a dedicated machine), firmware, microcode, or a combination thereof. In some embodiments, methods **400**A-D are performed, at least in part, by a controller (e.g., of control module **180** or controllers **185** of FIGS. **1**A-D, control module **94** of FIGS. **2**A-E, controller **303** of FIGS. **3**C-**3**N). In some embodiments, a non-transitory storage medium stores instructions that when executed by a processing device (e.g., of control module **180** or controllers **185** of FIGS. **1**A-D, control module **94** of FIGS. **2**A-E, controller **303** of FIGS. **3**C-**3**N), cause the processing device to perform one or more of methods **400**A-D.

(83) For simplicity of explanation, methods **400**A-D are depicted and described as a series of operations. However, operations in accordance with this disclosure can occur in various orders and/or concurrently and with other operations not presented and described herein. Furthermore, in some embodiments, not all illustrated operations are performed to implement methods **400**A-D in accordance with the disclosed subject matter. In addition, those skilled in the art will understand and appreciate that methods **400**A-D could alternatively be represented as a series of interrelated states via a state diagram or events.

(84) Referring to FIG. **4**A, at block **402**, processing logic (e.g., controller) determines (e.g., via user input) an internal lubrication operation of a pressure exchanger is to be performed.

(85) At block **404**, processing logic causes one or more first valves (e.g., fluidly coupled to one or more flushing outlets associated with circumferential grooves of the pressure exchanger) to be actuated to a closed position.

(86) At block **406**, processing logic causes a second valve (e.g., fluidly coupled to a lubrication outlet, located between lubrication outlet and one or more flushing inlets) to be actuated to an open position.

(87) The lubrication outlet may provide fluid flow (e.g., from HPOUT, from lubrication outlet) to one or more flushing inlets which enters (e.g., lubricates) radial bearing gap (e.g., via bearing fluid plenum), then flows to circumferential groove, then flows to axial bearing gaps, and then exits via a port of the pressure exchanger (e.g., low pressure out (LPOUT) port).

(88) Referring to FIG. **4**B, at block **422**, processing logic (e.g., controller) determines an external

lubrication operation of a pressure exchanger is to be performed.
(89) At block **424**, processing logic causes first valves (e.g., fluidly coupled to flushing outlets associated with circumferential grooves of the pressure exchanger) to be actuated to a closed position.
(90) At block **426**, processing logic causes a second valve (e.g., fluidly coupled to a lubrication outlet, located between lubrication outlet and flushing inlets) to be actuated to the closed position.
(91) At block **428**, processing logic causes external bearing fluid to be provided to the pressure exchanger (e.g., via flushing inlets associated with the radial bearing gap). The external bearing fluid may be provided to one or more flushing inlets which enters (e.g., lubricates) radial bearing gap (e.g., via bearing fluid plenum), then flows to circumferential groove, then flows to axial bearing gaps, and then exits via a port of the pressure exchanger (e.g., LPOUT port). Method **400**B of FIG. **4**B may be used when the process fluid (e.g., fluid exchanging pressure with another fluid in the pressure exchanger) is not suitable as a bearing fluid (e.g., the process fluid causes fouling and frequent stalls or slowdown of the rotor), the HPOUT bearing fluid supply can be shut off and high pressure external clean bearing fluid can be piped in to keep PX running for long times without fouling.
(92) Referring to FIG. **4**C, at block **442**, processing logic (e.g., controller) determines a flushing operation of a pressure exchanger is to be performed (e.g., while the pressure exchanger is not operating).
(93) At block **444**, processing logic causes first valves (e.g., fluidly coupled to flushing outlets associated with circumferential grooves of the pressure exchanger) to be actuated to an open position.
(94) At block **446**, processing logic causes a second valve (e.g., fluidly coupled to a lubrication outlet) to be actuated to a closed position.
(95) At block **448**, processing logic causes cleaning-in-place fluid to be provided to the pressure exchanger (e.g., via flushing inlets associated with the radial bearing gap). In method **400**C of FIG. **4**C, cleaning-in-place fluid (e.g., cleaning agent) may be provided to one or more flushing inlets which enters (e.g., flushes) radial bearing gap (e.g., via bearing fluid plenum), then flows to circumferential groove, and then exits via flushing outlets.
(96) Referring to FIG. **4**D, at block **462**, processing logic (e.g., controller) determines a flushing operation of a pressure exchanger is to be performed (e.g., while the pressure exchanger is operating).
(97) At block **464**, processing logic causes first valves (e.g., fluidly coupled to flushing outlets associated with circumferential grooves of the pressure exchanger) to be actuated to a closed position.
(98) At block **466**, processing logic causes a second valve (e.g., fluidly coupled to a lubrication outlet) to be actuated to an open position. In method **400**D of FIG. **4**D, lubrication outlet may provide fluid flow (e.g., from HPOUT) to a first flushing inlet which enters radial bearing gap and exits via a second flushing inlet (e.g., a valve disposed between second flushing inlet and lubrication outlet is in a closed position).
(99) FIG. **5** is a block diagram illustrating a computer system **500**, according to some embodiments. In some embodiments, the computer system **500** is a controller, controller device, client device, server, control module **180** or controllers **185** of FIGS. **1**A-D, control module **94** of FIGS. **2**A-E, controller **303** of FIGS. **3**C-N, etc.
(100) In some embodiments, computer system **500** is connected (e.g., via a network, such as a Local Area Network (LAN), an intranet, an extranet, or the Internet) to other computer systems. Computer system **500** operates in the capacity of a server or a client computer in a client-server environment, or as a peer computer in a peer-to-peer or distributed network environment. In some embodiments, computer system **500** is provided by a personal computer (PC), a tablet PC, a Set-Top Box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server,

a network router, switch or bridge, or any device capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that device. Further, the term "computer" shall include any collection of computers that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methods described herein.

(101) In some embodiments, the computer system **500** includes a processing device **502**, a volatile memory **504** (e.g., Random Access Memory (RAM)), a non-volatile memory **506** (e.g., Read-Only Memory (ROM) or Electrically-Erasable Programmable ROM (EEPROM)), and/or a data storage device **516**, which communicates with each other via a bus **508**.

(102) In some embodiments, processing device **502** is provided by one or more processors such as a general purpose processor (such as, for example, a Complex Instruction Set Computing (CISC) microprocessor, a Reduced Instruction Set Computing (RISC) microprocessor, a Very Long Instruction Word (VLIW) microprocessor, a microprocessor implementing other types of instruction sets, or a microprocessor implementing a combination of types of instruction sets) or a specialized processor (such as, for example, an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Digital Signal Processor (DSP), a PID controller, or a network processor). In some embodiments, processing device **502** is provided by one or more of a single processor, multiple processors, a single processor having multiple processing cores, and/or the like.

(103) In some embodiments, computer system **500** further includes a network interface device **522** (e.g., coupled to network **574**). In some embodiments, the computer system **500** includes one or more input/output (I/O) devices. In some embodiments, computer system **500** also includes a video display unit **510** (e.g., a liquid crystal display (LCD)), an alphanumeric input device **512** (e.g., a keyboard), a cursor control device **514** (e.g., a mouse), and/or a signal generation device **520**. Computer system **500** may include signal input device **515**, e.g., for receiving signals from other devices. For example, signal input device **515** may facilitate reception by computer system **500** of measurement data from sensors associated with a fluid handling system. Signal generate device **520** may be utilized to generate and/or send control signals for sending instructions to one or more components of a fluid handling system.

(104) In some implementations, data storage device **518** (e.g., disk drive storage, fixed and/or removable storage devices, fixed disk drive, removable memory card, optical storage, network attached storage (NAS), and/or storage area-network (SAN)) includes a non-transitory computer-readable storage medium **524** on which stores instructions **526** encoding any one or more of the methods or functions described herein, and for implementing methods described herein. Control module **527** (e.g., including any of controllers and/or control modules of the present disclosure) may be included in instructions **526**.

(105) In some embodiments, instructions **526** also reside, completely or partially, within volatile memory **504** and/or within processing device **502** during execution thereof by computer system **500**, hence, volatile memory **504** and processing device **502** also constitute machine-readable storage media, in some embodiments.

(106) While computer-readable storage medium **524** is shown in the illustrative examples as a single medium, the term "computer-readable storage medium" shall include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of executable instructions. The term "computer-readable storage medium" shall also include any tangible medium that is capable of storing or encoding a set of instructions for execution by a computer that cause the computer to perform any one or more of the methods described herein. The term "computer-readable storage medium" shall include, but not be limited to, solid-state memories, optical media, and magnetic media.

(107) The methods, components, and features described herein may be implemented by discrete hardware components or may be integrated in the functionality of other hardware components such as ASICS, FPGAs, DSPs, or similar devices. In addition, the methods, components, and features

may be implemented by firmware modules or functional circuitry within hardware devices. Further, the methods, components, and features may be implemented in any combination of hardware devices and computer program components, or in computer programs.

(108) Unless specifically stated otherwise or clear from context, terms such as “receiving,” “transmitting,” “determining,” “generating,” “causing,” “actuating,” “adjusting,” “controlling,” “identifying,” “providing,” or the like, refer to actions and processes performed or implemented by computer systems that manipulates and transforms data represented as physical (electronic) quantities within the computer system registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices. Also, the terms “first,” “second,” “third,” “fourth,” etc. as used herein are meant as labels to distinguish among different elements and may not have an ordinal meaning according to their numerical designation.

(109) Examples described herein also relate to an apparatus for performing the methods described herein. This apparatus may be specially constructed for performing the methods described herein, or it may include a general purpose computer system selectively programmed by a computer program stored in the computer system. Such a computer program may be stored in a computer-readable tangible storage medium.

(110) The methods and illustrative examples described herein are not inherently related to any particular computer or other apparatus. Various general purpose systems may be used in accordance with the teachings described herein, or it may prove convenient to construct more specialized apparatus to perform methods described herein and/or each of their individual functions, routines, subroutines, or operations. Examples of the structure for a variety of these systems are set forth in the description above.

(111) Although the operations of the methods herein are shown and described in a particular order, the order of the operations of each method may be altered so that certain operations may be performed in an inverse order or so that certain operation may be performed, at least in part, concurrently with other operations. In another embodiment, instructions or sub-operations of distinct operations may be in an intermittent and/or alternating manner.

(112) The preceding description sets forth numerous specific details, such as examples of specific systems, components, methods, and so forth, in order to provide a good understanding of several embodiments of the present disclosure. It will be apparent to one skilled in the art, however, that at least some embodiments of the present disclosure may be practiced without these specific details. In other instances, well-known components or methods are not described in detail or are presented in simple block diagram format in order to avoid unnecessarily obscuring the present disclosure. Thus, the specific details set forth are merely exemplary. Particular implementations may vary from these exemplary details and still be contemplated to be within the scope of the present disclosure. Descriptions of systems herein may include descriptions of one or more optional components. Components may be included in combinations not specifically discussed in this disclosure, and still be within the scope of this disclosure.

(113) Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrase “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. In addition, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” When the term “about,” “substantially,” or “approximately” is used herein, this is intended to mean that the nominal value presented is precise within ±10%. Also, the terms “first,” “second,” “third,” “fourth,” etc. as used herein are meant as labels to distinguish among different elements and can not necessarily have an ordinal meaning according to their numerical designation.

(114) The terms “over,” “under,” “between,” “disposed on,” “before,” “after,” and “on” as used

herein refer to a relative position of one material layer or component with respect to other layers or components. For example, one layer disposed on, over, or under another layer may be directly in contact with the other layer or may have one or more intervening layers. Moreover, one layer disposed between two layers may be directly in contact with the two layers or may have one or more intervening layers. Similarly, unless explicitly stated otherwise, one feature disposed between two features may be in direct contact with the adjacent features or may have one or more intervening layers or components.
(115) It is to be understood that the above description is intended to be illustrative, and not restrictive. Many other embodiments will be apparent to those of skill in the art upon reading and understanding the above description. The scope of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which each claim is entitled.

Claims

1. A system comprising: a pressure exchanger comprising: a rotor configured to exchange pressure between a first fluid and a second fluid; a housing disposed around the rotor; one or more flushing inlets coupled to the housing; and one or more flushing outlets coupled to the housing; and one or more first valves coupled to the one or more flushing outlets, wherein the one or more first valves in an open position are associated with a flushing operation, and wherein the one or more first valves in a closed position are associated with a lubrication operation.
2. The system of claim 1 further comprising a lubrication outlet and a second valve coupled to the lubrication outlet, the second valve having an open position and a closed position, wherein: the one or more first valves in the closed position and the second valve in the closed position are associated with the lubrication operation being an external lubrication operation; the one or more first valves in the closed position and the second valve in the open position are associated with the lubrication operation being an internal lubrication operation; and the one or more first valves in the open position and the second valve in the closed position are associated with the flushing operation.
3. The system of claim 2, wherein the lubrication outlet is fluidly coupled to a high pressure outlet port of the pressure exchanger.
4. The system of claim 2 further comprising a third valve having an open position and a closed position, the third valve being associated with at least one of the one or more flushing inlets, wherein: the one or more first valves in the closed position, the second valve in the open position, and the third valve in the closed position are associated with lubrication fluid entering through a first flushing inlet of the one or more flushing inlets and exiting through a second flushing inlet of the one or more flushing inlets to a drainage line.
5. The system of claim 1, wherein the one or more flushing inlets are fluidly coupled with a radial bearing gap between the rotor and a sleeve disposed around the rotor.
6. The system of claim 1, wherein the one or more flushing outlets are fluidly coupled with circumferential grooves between the rotor and end covers disposed at distal ends of the rotor.
7. The system of claim 1, wherein at least one of the one or more flushing inlets or the one or more flushing outlets are associated with flushing one or more of: a radial bearing gap between the rotor and a sleeve of the pressure exchanger; circumferential grooves between the rotor and end covers disposed at distal ends of the rotor; an axial bearing gap disposed between the rotor and the end covers; a sleeve outside diameter disposed between the sleeve and the housing; or a tension rod disposed in the rotor.
8. The system of claim 1, wherein: an inner surface of a sleeve of the pressure exchanger forms a collection groove; the sleeve is disposed around the rotor; the rotor forms a channel between a duct and an outside surface of the rotor; and particles are to be collected in the collection groove and flushed out through the duct responsive to the channel aligning with the collection groove.

9. The system of claim 1, wherein: a sleeve is disposed around the rotor; an outer surface of the rotor forms a collection groove; and the sleeve forms a passage that connects the collection groove with an outlet of the pressure exchanger responsive to the collection groove aligning with the passage.
10. The system of claim 1 further comprising an end cap disposed at a distal end of the pressure exchanger and a bearing fluid filter disposed in the end cap, wherein the pressure exchanger is to direct a portion of the first fluid or the second fluid exiting the pressure exchanger through the bearing fluid filter to bearings of the pressure exchanger.
11. The system of claim 1 further comprising a coating on one or more of: an end-cover circumferential groove; a rotor center bore; a rotor duct; a sleeve outer surface; a rotor outer surface; or a rotor axial surface.
12. The system of claim 11, wherein the coating is one or more of a hydrophobic coating, anti-adhesion coating, oleophobic coating, or silver nano particle coating.
13. A pressure exchanger comprising: a rotor configured to exchange pressure between a first fluid and a second fluid; a housing disposed around the rotor; one or more flushing inlets coupled to the housing and associated with a radial bearing gap of the pressure exchanger; and flushing outlets coupled to the housing and associated with circumferential grooves of the pressure exchanger, wherein one or more first valves are coupled to the flushing outlets, wherein the one or more first valves in an open position are associated with a flushing operation, and wherein the one or more first valves in a closed position are associated with a lubrication operation.
14. The pressure exchanger of claim 13 further comprising a lubrication outlet and a second valve coupled to the lubrication outlet, the second valve having an open position and a closed position, wherein: the one or more first valves and the second valve in the closed position are associated with an external lubrication operation; the one or more first valves in the closed position and the second valve in the open position are associated with an internal lubrication operation; and the one or more first valves in the open position and the second valve in the closed position are associated with the flushing operation.
15. The pressure exchanger of claim 13, wherein: a sleeve of the pressure exchanger is disposed around the rotor; an inner surface of the sleeve forms a collection groove; the rotor forms a channel between a duct and an outside surface of the rotor; and particles are to be collected in the collection groove and flushed out through the duct responsive to the channel aligning with the collection groove.
16. The pressure exchanger of claim 13, wherein: a sleeve of the pressure exchanger is disposed around the rotor; an outer surface of the rotor forms a collection groove; and the sleeve forms a passage that connects the collection groove with a low pressure outlet of the pressure exchanger responsive to the collection groove aligning with the passage.
17. The pressure exchanger of claim 13 further comprising an end cap disposed at a distal end of the pressure exchanger and a bearing fluid filter disposed in the end cap, wherein the pressure exchanger is to direct a portion of fluid exiting the pressure exchanger through the bearing fluid filter to bearings of the pressure exchanger.
18. A method comprising: causing one or more first valves to be actuated to an open position, the one or more first valves being coupled to flushing outlets of a pressure exchanger, wherein the flushing outlets are coupled to a housing of the pressure exchanger and are associated with circumferential grooves of the pressure exchanger; and causing a second valve to be actuated to a closed position, the second valve being associated with a lubrication outlet, wherein the one or more first valves in the open position and the second valve in the closed position are associated with a flushing operation of the pressure exchanger.
19. The method of claim 18 further comprising causing a cleaning-in-place fluid to flow via one or more flushing inlets coupled to the housing and associated with a radial bearing gap of the pressure exchanger to cause fluid flow through the radial bearing gap to the circumferential grooves and out

via the flushing outlets.
20. The method of claim 18 further comprising causing the one or more first valves to be actuated to a closed position and the second valve to be actuated to an open position to perform an internal lubrication operation of the pressure exchanger.